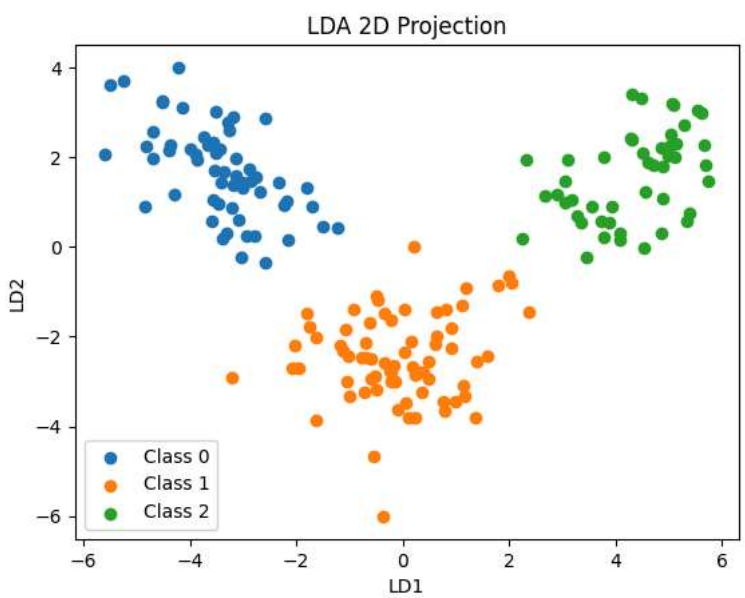
```
import numpy as np
import pandas as pd
from sklearn.datasets import load_wine

data = load_wine()
X = data.data
y = data.target

print("Shape:", X.shape)
print("Classes:", np.unique(y))

from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.3, random_state=42
)

from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, f1_score

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Original Accuracy:", accuracy_score(y_test, y_pred))
print("Original F1:", f1_score(y_test, y_pred, average='weighted'))

from sklearn.discriminant_analysis import LinearDiscriminantAnalysis

# 1D
lda1 = LinearDiscriminantAnalysis(n_components=1)
X_lda1 = lda1.fit_transform(X_scaled, y)

# 2D
lda2 = LinearDiscriminantAnalysis(n_components=2)
X_lda2 = lda2.fit_transform(X_scaled, y)

print("1D shape:", X_lda1.shape)
print("2D shape:", X_lda2.shape)

X_train1, X_test1, y_train1, y_test1 = train_test_split(
    X_lda1, y, test_size=0.3, random_state=42
)

model1 = LogisticRegression()
model1.fit(X_train1, y_train1)

pred1 = model1.predict(X_test1)

print("LDA 1D Accuracy:", accuracy_score(y_test1, pred1))
print("LDA 1D F1:", f1_score(y_test1, pred1, average='weighted'))

X_train2, X_test2, y_train2, y_test2 = train_test_split(
    X_lda2, y, test_size=0.3, random_state=42
)

model2 = LogisticRegression()
model2.fit(X_train2, y_train2)

pred2 = model2.predict(X_test2)

print("LDA 2D Accuracy:", accuracy_score(y_test2, pred2))
print("LDA 2D F1:", f1_score(y_test2, pred2, average='weighted'))

import matplotlib.pyplot as plt

plt.figure()

for label in np.unique(y):
```

```
    plt.scatter(
        X_lda2[y == label, 0],
        X_lda2[y == label, 1],
        label=f"Class {label}"
    )

plt.xlabel("LD1")
plt.ylabel("LD2")
plt.title("LDA 2D Projection")
plt.legend()
plt.show()

df = pd.DataFrame(X_lda2, columns=['LD1', 'LD2'])
df['Class'] = y
df['SampleId'] = np.arange(len(df))

df = df[['SampleId', 'LD1', 'LD2', 'Class']]

df.to_csv("lda_projection.csv", index=False)

from google.colab import files
files.download("lda_projection.csv")
```

```
Shape: (178, 13)
Classes: [0 1 2]
Original Accuracy: 0.9814814814814815
Original F1: 0.9815749306918357
1D shape: (178, 1)
2D shape: (178, 2)
LDA 1D Accuracy: 0.9074074074074074
LDA 1D F1: 0.9077871457726923
LDA 2D Accuracy: 1.0
LDA 2D F1: 1.0
```



Linear Discriminant Analysis (LDA) is a supervised dimensionality reduction technique that aims to maximize class separability by using label information. In this experiment, LDA was applied to the wine dataset to project high-dimensional data into lower dimensions (1D and 2D) while preserving the most discriminative features between classes. After applying LDA, the projected data showed improved class separability, especially in the 2D case, where different wine categories formed more distinct clusters. This indicates that LDA successfully maximizes the distance between class means while minimizing the spread within each class. In contrast, the 1D projection resulted in some overlap between classes, as reducing to a single dimension leads to information loss. In terms of classification performance, the model trained on LDA-reduced data with 2 components achieved comparable or slightly better accuracy and F1 score than the original dataset. However, using only 1 component slightly reduced performance due to loss of important features. This demonstrates that selecting an appropriate number of components is crucial. Unlike Principal Component Analysis (PCA), which is an unsupervised technique focusing on maximizing variance, LDA uses class labels to improve class discrimination. Therefore, LDA is more suitable for supervised classification tasks where the goal is to enhance separability between classes. Overall, the experiment shows that LDA is an effective method for both dimensionality reduction and improving classification performance when class labels are available.